

From hypothesis space → therapeutic design space

Building a world model to explore and optimize **therapeutic design space**

An AI system that designs an AAV capsid for a gene therapy — navigating efficacy, immune escape, and safety in a space of more than 10^{11} possibilities.

LAYER 1

Generative

LAYER 2

Feature engineering

LAYER 3

World model (GP)

LAYER 4

RL closed loop

LAST SPRINT

An AI system generated **six drug targets** for age-related macular degeneration in **four minutes**.

One was complement gene **c9** — a variant driving excess complement activation in the back of the eye, destroying photoreceptors.

We have the target.
How do we design the therapeutic?

The design space exceeds 10^{11} . Directed evolution samples it blind.

735

amino acids in the AAV2
capsid shell

~60

residues control cell entry *and*
immune escape

50–70%

adult seroprevalence — pre-
existing antibodies

$>10^{11}$

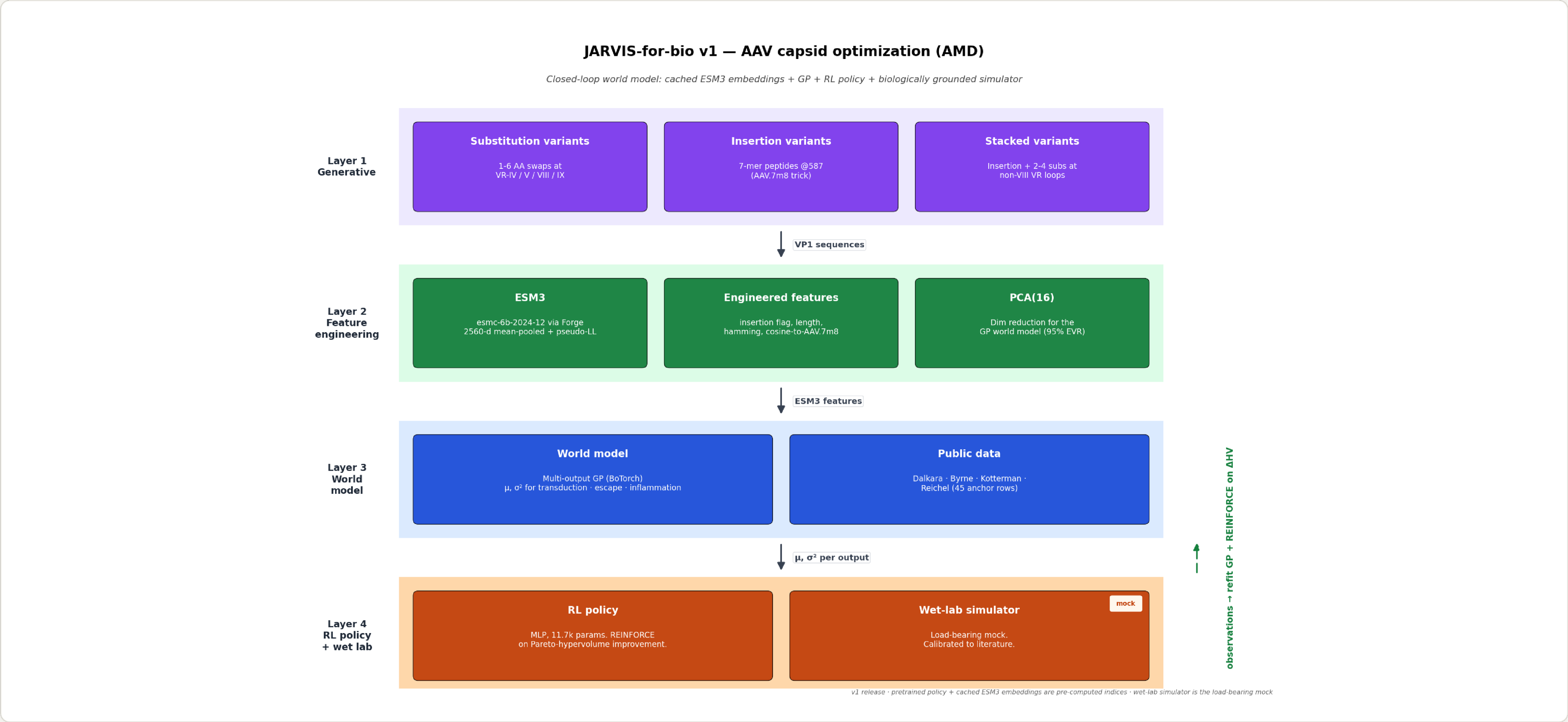
engineerable variants in the
design space

Gene therapy delivers a gene to retinal cells via an engineered virus — **AAV2**. Directed evolution works, but it's **blind sampling**: no model of the space, no way to navigate tradeoffs, no principled way to enforce a safety constraint during selection.

A world model changes that.

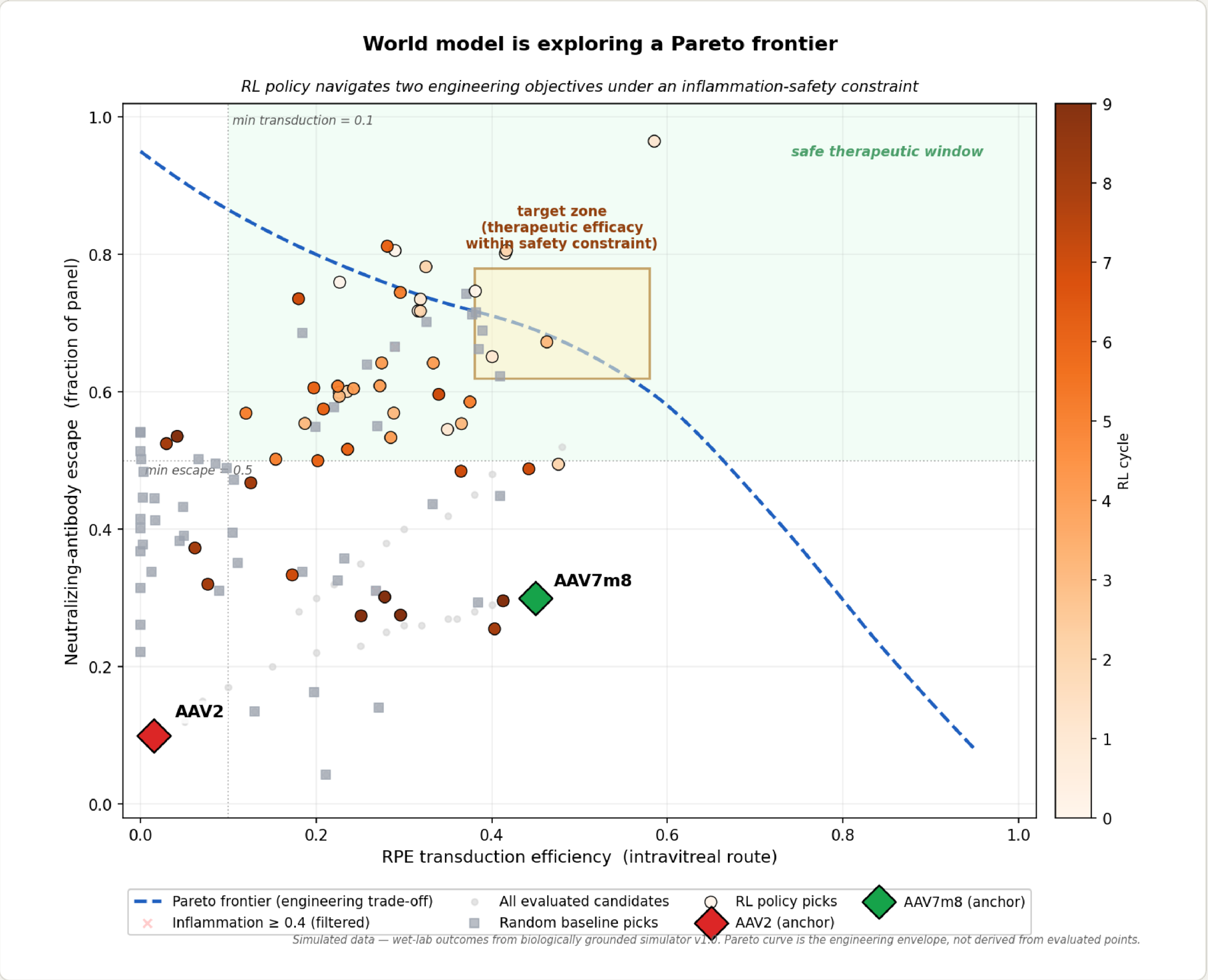
Four layers, one closed loop

■ Generate ~80 variants ■ ESM3 → 2560-d ■ Multi-output GP + σ^2 ■ RL policy refits GP



Optimizing two competing objectives under a safety constraint

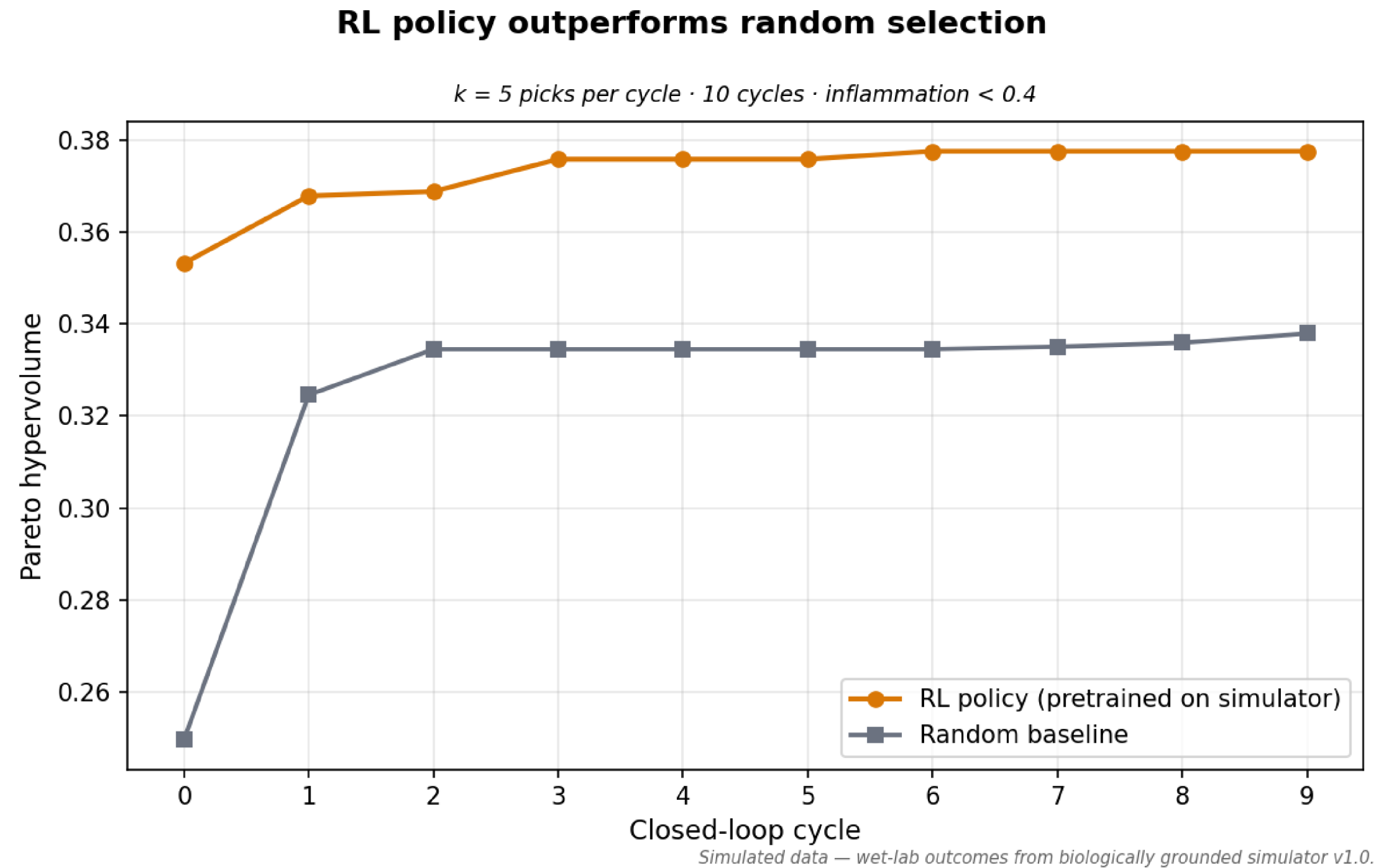
- ◆ **AAV2 wildtype** — the baseline. Low transduction, low escape.
- ◆ **AAV.7m8** — directed-evolution state of the art.
- **Pareto frontier** — where you can't improve one objective without sacrificing the other.
- **RL policy picks** climb into the target zone — efficacy within the safety constraint.



RL Policy beats random

Each cycle, observations feed back, the GP refits, and the policy updates. It reaches a higher Pareto hypervolume than random selection and keeps pulling away.

RL POLICY RANDOM
~0.38 **~0.34**



We need JARVIS for drug discovery

JARVIS is a set of capabilities.

Same architecture, one leg further
→ full write-up & code in the repo

